

Mapping Vegetation Patterns in Jim Corbett National Park Using NDVI: An Introductory Remote Sensing Study

Author: Siddhant Katariya

Software: QGIS, Copernicus Browser

Data Source: Sentinel-2 Satellite Imagery

1. Introduction

I came up with this Project when I wanted to create my first one after learning about GIS and Remote Sensing. This is the first project I have made. The core of this project is the **Normalized Difference Vegetation Index (NDVI)**, which estimates vegetation density and condition by comparing the amount of red light absorbed by plants with the amount of near-infrared light reflected by healthy vegetation.

The aim of this project was not to produce new ecological findings but to develop a practical understanding of the complete remote sensing process using QGIS and Sentinel-2 imagery. The project focused on generating an NDVI map for a selected area within Jim Corbett National Park and interpreting the observed vegetation patterns while recognising the limitations of the method.

2. Study Area

The study area is located within Jim Corbett National Park in Uttarakhand, India. A section of the park containing river channels, the dam's reservoir

and its shoreline, hills, and forested regions, selected because it provides a variety of landscape features suitable for observing differences in vegetation.

The combination of water bodies, relatively flat terrain, and elevated areas made the location appropriate for learning how vegetation indices respond to different land surface conditions.

Figure 1. Study area in true-colour Sentinel-2 imagery.

3. Data and Methods

Data

- Sentinel-2 satellite imagery
- Bands used:
 - Band 4 (Red)
 - Band 8 (Near Infrared)

Software

- QGIS
- Copernicus Browser

Methodology

The following workflow was followed:

1. A cloud-free Sentinel-2 image covering the study area was selected using the Copernicus Browser.
2. Band 4 (Red) and Band 8 (Near Infrared) were downloaded as GeoTIFF files.
3. Both bands were imported into QGIS.
4. The Raster Calculator was used to calculate NDVI using the standard equation:

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

5. The resulting NDVI raster was classified using a pseudocolour colour ramp to improve visual interpretation.
 6. Vegetation patterns were examined by comparing NDVI values across rivers, shoreline areas, hills, and interior forest regions.
-

4. Results

The generated NDVI map showed noticeable spatial variation in vegetation density throughout the study area.

Several observations were made:

- River edges generally exhibited relatively high NDVI values.
- Vegetation distribution across flatter terrain was patchy rather than continuous.
- Interior regions did not consistently display the highest vegetation density, contrary to the initial expectation.
- Some hill-adjacent areas also exhibited relatively high NDVI values.

These observations suggest that vegetation distribution within the selected study area is spatially heterogeneous rather than uniform.

Figure 2. Final NDVI map generated in QGIS.

5. Discussion

Prior to the analysis, it was expected that the central portions of the study area would consistently contain the densest vegetation. However, the NDVI map demonstrated that vegetation patterns were considerably more variable.

Higher NDVI values frequently occurred near river margins, indicating that proximity to water may influence vegetation growth in some locations. At the same time, numerous isolated patches of high vegetation density were observed away from rivers, suggesting that additional environmental factors such as local topography, soil conditions, canopy structure, or species composition may also influence vegetation distribution.

This project also demonstrated that satellite imagery should not be interpreted as a conventional photograph. Instead, each pixel represents measured spectral information that can be processed mathematically to reveal environmental characteristics not immediately visible in true-colour imagery.

6. Limitations

Several limitations should be considered when interpreting the results:

- Only a single satellite image was analysed, so seasonal variation was not considered.
 - NDVI indicates vegetation presence and relative vigour but cannot determine the underlying ecological causes of observed patterns.
 - Mixed pixels and canopy effects may influence NDVI values.
 - The spatial resolution of Sentinel-2 limits the level of detail that can be observed at smaller scales.
-

7. Conclusion

This project successfully introduced the complete workflow of a basic remote sensing analysis using Sentinel-2 imagery and QGIS. An NDVI map was generated from raw satellite bands and used to investigate vegetation patterns within a selected area of Jim Corbett National Park.

Although the project was exploratory, it provided valuable practical experience in handling satellite imagery, calculating vegetation indices, interpreting spatial patterns, and recognising the strengths and limitations of NDVI. The experience also highlighted the importance of approaching environmental interpretation cautiously, allowing observations to guide conclusions rather than relying solely on initial assumptions.

8. Personal Reflection

This project marked my first complete remote sensing workflow. Before beginning, I viewed satellite imagery primarily as photographs. Through this analysis, I learned that satellite images are collections of spectral measurements that can be processed to reveal environmental information beyond what is visible to the human eye.

One of the most surprising outcomes was the patchy distribution of vegetation across the study area. My initial assumption was that vegetation density would increase toward the central regions of the forest; however, the analysis showed that the landscape was considerably more complex. This experience reinforced the importance of allowing evidence to challenge preconceived expectations.

Beyond learning the technical aspects of QGIS and NDVI, this project strengthened my interest in applying GIS and remote sensing to investigate environmental questions. It also established a foundation for future projects involving urban vegetation, land surface temperature, and environmental monitoring.
